1 a) $a = (2, 1, 3)$

$$P = \frac{a a^T}{a^T a} = \frac{1}{14}\begin{bmatrix} 4 & 2 & 6 \\ 2 & 1 & 3 \\ 6 & 3 & 9 \end{bmatrix}$$

b) $C(P) = c\, a$ (line)

$$N(P) = \begin{bmatrix} 4 & 2 & 6 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= c_1 \begin{bmatrix} -3/2 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix} \text{ (plane)}$$

c) $\lambda = 0, 0, 1$

$$\lambda_1 = \begin{bmatrix} -3/2 \\ 0 \\ 1 \end{bmatrix} \quad \lambda_2 = \begin{bmatrix} -1/2 \\ 1 \\ 0 \end{bmatrix} \quad \lambda_3 = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$

(2) a) $p = A\hat{x}$

$$Ax = b$$

$$p$$

$$A^T A \hat{x} = (A^T b)$$

$$\boxed{\hat{x} = (A^T A)^{-1} A^T b}$$

$$e = (I - A(A^T A)^{-1} A^T)\, b$$

$e \cdot v = 0$

$$\boxed{v = C(A)}$$

\# $A^T A$ is not invertible when A has dependent columns.

b) $A = QR$

$(QR)x = b$

$\hat{x} = (A^T A)^{-1} A^T$

$p = (QR)(R^T Q^T Q R)^{-1} R^T Q^T$

$p = QR \left(R^T R\right)^{-1} (QR)^T b$

$= Q\cancel{R}\cancel{R^{-1}} \cancel{(R^T)^{-1} R^T} Q^T = QQ^T b$

c) $q_1 \; q_2$ ortho normal $\mathbb{R}^5$

$\Rightarrow \quad Q = \begin{bmatrix} | & | \\ q_1 & q_2 \\ | & | \end{bmatrix}$

$p = (q_1 q_1^T + q_2 q_2^T) v$

③ $A_n = \begin{bmatrix} 0 & -1 & -1 & \cdots \\ -1 & 0 & -1 & \cdots \\ & & 0 & -1 \\ -1 & -1 & 0 & \\ -1 & -1 & -1 & 0 \end{bmatrix}$

a) $\det A_n$

$A_2 = \begin{vmatrix} 0 & -1 \\ -1 & 0 \end{vmatrix}$

$\det A_2 = -1$

$$\begin{bmatrix} 0 & -1 & -1 & \cdots \\ -1 & 0 & -1 & \\ -1 & -1 & 0 & \\ -1 & -1 & \cdots & 0 \end{bmatrix}$$

$$\det \begin{bmatrix} 0 & -1 & -1 & \cdots \\ \vdots & & & \\ -(n-1) & -(n-1) & \cdots & -(n-1) \end{bmatrix}$$

$$D_n = -(n-1)\left(-D_n + D_{n-1}\right)$$

$$D_n = (n-1) D_n - (n-1) D_{n-1}$$

$$D_n (1-n+1) = (1-n) D_{n-1}$$

$$\boxed{D_n = \frac{1-n}{2-n} D_{n-1}}$$

$$D_n = \left(\frac{n-1}{n-2}\right) D_{n-1}$$

$$D_n = \frac{(n-1)(n-2)\ \dots\ (2)}{(n-2)\ -\ -\ -\ 1}$$

$$\frac{(n-1)!}{(n-2)!} \qquad \cancel{A}_2$$

$$(n-1)(-1)$$

$$\boxed{(1-n)}$$

b) $A = \begin{bmatrix} 0 & -1 & -1 & - - \\ -1 & 0 & -1 & \\ -1 & & & \\ \vdots & & & \end{bmatrix}$

$A = -1 \begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & & 0 \\ 1 & & & & \end{bmatrix}$

c) $A_3 x = x$

$$\begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{bmatrix} x = (1)\, x$$

$$\boxed{\lambda = 1 \quad , \quad 1 \quad , \quad -2} \rightarrow \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & -1 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & \cdot \end{bmatrix} \qquad \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$